OmniShow: Unifying Multimodal Conditions for Human-Object Interaction Video Generation

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Research Questions

- How to unify text, reference images, audio, and pose conditions for human-object interaction video generation without compromising generation quality?
- How to efficiently leverage heterogeneous sub-task datasets (R2V, A2V) to train a unified multimodal video generation model?
- How to achieve precise audio-visual synchronization while maintaining controllability across multiple modalities?

Core Idea

Existing video generation methods handle only subsets of conditions (text+images, text+audio, etc.) but no unified framework exists for human-object interaction videos requiring all four modalities. OmniShow's key insight is a divide-and-conquer approach: (1) Use channel concatenation to inject image/pose conditions via 'pseudo-frame tokens' that preserve the base model's token structure, (2) Design gated attention that restricts video tokens to interact only with corresponding audio segments for temporal alignment, (3) Train specialized R2V and A2V models separately, then merge via weight interpolation to create emergent multimodal capabilities. This avoids disrupting pretrained generative priors while enabling precise multimodal control.

Key Formula

Unified Channel Concatenation

$$x_{in} = \mathrm{Concat}([x', \|, x_t], [r\|, p], [m'\|, m])$$

x' are pseudo-frame tokens, x_t are noisy video tokens, r are reference image tokens, p are pose tokens, m' and m are corresponding binary masks. This extends the original model input by concatenating pseudo-frames with actual video tokens along temporal dimension while preserving channel structure.

Gated Local-Context Attention

$$\mathrm{Attn}(Q, K, V, M) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + \log M\right)V$$

Q derived from video tokens, K,V from audio tokens, M is binary mask restricting video tokens to interact only with corresponding audio segments. The log M term enforces temporal alignment by masking out non-corresponding audio-video pairs for precise synchronization.

Key Figure

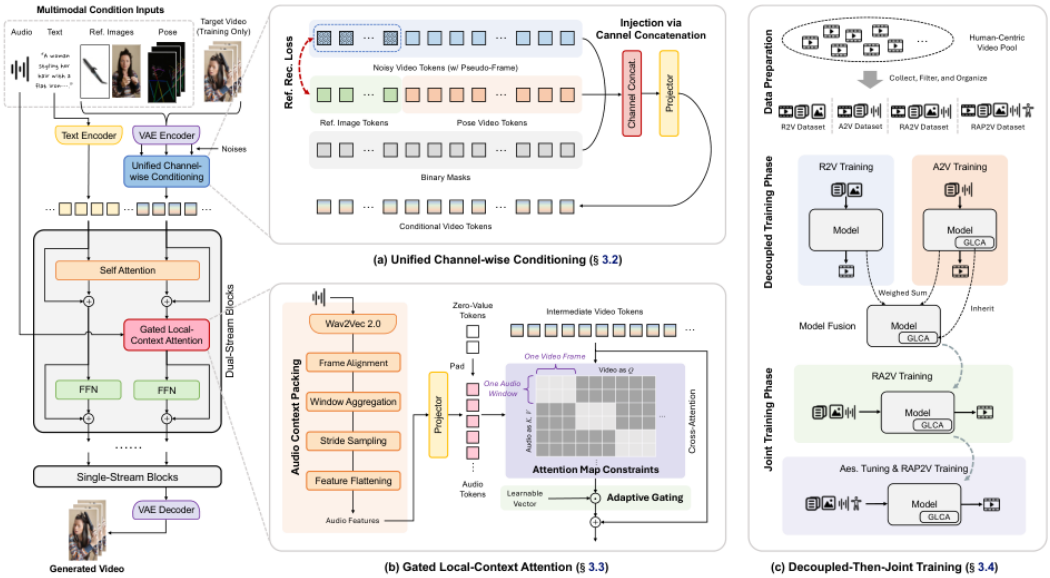

Figure 2 Pipeline of OmniShow. Our framework consists of: *(a) Unified Channel-wise Conditioning* (section 3.2) effectively injects reference images and pose cues via unified channel concatenation. It augments noisy video tokens with pseudo-frames, which are supervised by a reference reconstruction loss to preserve semantic details. *(b) Gated Local-Context Attention* (section 3.3) ensures precise audio-visual synchronization. It packs raw audio features with sufficient contextual information and injects them via masked attention to align video frames with corresponding audio segments, followed by adaptive gating to stabilize early training. *(c) Decoupled-Then-Joint Training* (section 3.4) makes the efficient utilization of heterogeneous datasets possible. We first train specialized R2V and A2V models on separate sub-task datasets, then fuse them via weight interpolation, followed by joint fine-tuning to unify multimodal capabilities. Note that we omit some components, such as normalization, within the MMDiT blocks for brevity.

Figure 2: Complete architecture overview showing the three core components: Unified Channel-wise Conditioning, Gated Local-Context Attention, and Decoupled-Then-Joint Training pipeline.

Limitations

- Current evaluation focuses on 5-second clips, limiting assessment of longer video generation capabilities
- Human reference images in benchmark are AI-generated, potentially introducing distribution biases compared to real photos
- Model requires separate training phases (R2V, A2V, then joint) increasing training complexity and computational cost
- Performance may degrade in extreme scenarios with conflicting multimodal inputs or overly intense motion